D 11760 **(Pages : 3)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION
NOVEMBER 2021

B.Com.

BCM 3A 11—BASIC NUMERICAL SKILLS

(2017–2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each question carries 1 *mark.*

I. Choose the correct answer :

1 Which of the following are examples of null set ?

(a) Set of even prime numbers.

(b) Set of letters in English alphabets.

(c) Set of odd natural numbers divisible by 2.

(d) All the above.

2 A matrix in which every element is Zero :

(a) Unit. (b) Diagonal.

(c) Scalar. (d) Null.

3 Equation ax2 + b = 0

(a) Pure quadratic equation.

(b) General quadratic equation.

(c) Not a quadratic equation.

(d) None.

4 A series obtained by adding a constant number to its preceding terms is

(a) GP. (b) AP.

(c) GP or AP. (d) None.

5 Ogives can be used to locate

(a) Median. (b) Quartiles.

(c) Deciles. (d) All.

Turn over

II. Fill in the blanks.

6. Find the 10th term of the series 4, 2, 0, –2 ————————.
7. The transpose of a matrix A is denoted by ————————.
8. $x^2 - 4 = 0$ implies x = ————————..
9. The number of observations corresponding to a particular class is known as ————————.
10. Mean of 3 items is 30 two of them are 20 and 30. The third number is ————————.

(10 × 1 = 10 marks)

Part B

Answer any eight *questions.*
Each question carries 2 marks.

11. What is a power set ?
12. If A = {1, 3, 5, 7 }, B = { 5, 9, 13, 17 } C = {1, 3, 9, 13 }, find :

    (a) A ∩ B ; (b) (A - B) - C.
13. Solve $7x - 21 - 3x + 13 = 7 + 6x - 19$.
14. Find two numbers whose sum is 30 and difference is 4.
15. What is an arithmetic progression ?
16. Find the 12th term of 2, 6, –18, 54 ————————.
17. What is sampling ? What are the two methods or techniques of sampling ?
18. Find the median for the following data : 4, 25, 45, 15, 26, 35, 55, 28, 48.
19. What are the absolute measures of dispersion or variability of data ?
20. Find the simple interest and amount for Rs. 25,000 at 10% p. a for 26 weeks. Assume 52 weeks in a year.

(8 × 2 = 16 marks)

Part C

Answer any **six** *questions.*
Each question carries 4 *marks.*

21. Each student in a class, studies at least one of the subject English, Mathematics and Accountancy. 16 study English, 22 Accountancy and 26 Mathematics. 5 study English and Accountancy, 14 study Mathematics and Accountancy and 2 English, Accountancy and Mathematics. Find the number of student who study

    (i) English and Mathematics; (ii) English, Mathematics but not Accountancy.

22 Solve the equation :

$$2\begin{bmatrix} x & y \\ z & t \end{bmatrix} + 3\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 5\begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}.$$

23 Solve $4x + 2y = 6$ and $5x + y = 6$.

24 Which term of the A.P 21, 18, 15, ———————— is - 81 (negative eighty- one) ?

25 X borrowed Rs. 26,400 from a bank to buy a scooter at the rate of 15% p.a. compounded yearly. What amount will be pay at the end of 2 years and 4 months to clear the loan ?

26 Draw a multiple ba r diagram for the following data :

	Production [in units]		
Year	A	B	C
2000	45	55	65
2009	35	60	70
2010	50	70	80
2011	55	80	60

27. What is kurtosis ? What are its types ?

28. Compete Standard Deviation of the series 4, 8, 10, 12, 15, 9, 7, 7.

(6 × 4 = 24 marks)

Part D

Answer any **two** *questions.*
Each question carries 15 *marks.*

29. Find the sum of all natural numbers from 1 to 200 excluding those divisible by 5.

30. If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, then verify that $(AB)^{-1} = B^{-1} A^{-1}$.

31. What is Index numbers ? What are its uses? What are the weighted aggregate methods of calculating index and give equations ?

(2 × 15 = 30 marks)